AirHunt: Bridging VLM Semantics and Continuous Planning for Efficient Aerial Object Navigation

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***Abstract*—Recent advances in large Vision-Language Models (VLMs) provide rich semantic understanding, enabling drones to search for open-set objects via natural language instructions. However, prior systems struggle to integrate VLMs into aerial platforms due to the orders-of-magnitude frequency mismatch between VLM inference and real-time planning, VLMs' limited 3D scene understanding, and the lack of a unified mechanism balancing semantic guidance with motion efficiency in large-scale environments. We present `AirHunt`, an aerial object navigation system that efficiently locates open-set objects with zero-shot generalization in outdoor environments. `AirHunt` features a dual-pathway asynchronous architecture that establishes a synergistic interface between VLM reasoning and path planning, enabling continuous flight with adaptive semantic guidance that evolves through motion. We further propose an active dual-task reasoning module that exploits geometric and semantic redundancy for selective VLM querying, and a semantic-geometric coherent planning module that dynamically reconciles semantic priorities with motion efficiency, enabling seamless adaptation to environmental heterogeneity. In simulation, `AirHunt` raises success rate by 49.1 percentage points over the strongest baseline and shortens flight time by 2.5×, while real-world experiments further validate its practical capability in complex outdoor scenes.**

***Index Terms*—Perception and autonomy, motion and path planning, vision-based navigation.**

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The code will be released at https://github.com/Robotics-STAR-Lab/AirHunt

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I. Introduction

AUTONOMOUS target search underpins critical applications such as disaster response, aerial surveillance, and environmental surveying. However, conventional search strategies remain limited by closed-set detectors and geometry-driven heuristics [1]. The emergence of Vision-Language Models (VLMs) has shifted this paradigm toward open-vocabulary semantic object navigation, enabling drones to interpret language instructions and leverage environmental context to improve search efficiency in complex environments. Despite this promise, integrating VLMs into aerial systems remains challenging in practice.

The first challenge stems from existing VLM-based search system designs, which treat VLMs as step-wise planners generating discrete actions [2], [3] and thus introduce two fundamental bottlenecks. (*i*) *Frequency mismatch:* the high inference latency of VLMs (often exceeding 2000ms [4]) restricts them to sparse action outputs that cannot match drones' real-time planning (>10 Hz). The resulting "stop-and-go" behavior forces drones to hover for VLM responses, disrupting continuous flight and raising failure risk under limited endurance. (*ii*) *Limited 3D scene understanding:* inferring from 2D observations, VLMs struggle to associate visual entities across viewpoints into a coherent global context, causing premature abandonment of promising regions and redundant revisits during long outdoor missions.

Secondly, existing planning strategies lack a unified framework that dynamically balances VLM-derived semantic guidance with geometric motion efficiency [2], [5]. Efficient aerial object navigation requires exploiting semantic cues and considering motion costs, yet current methods mostly adopt greedy decisions or simplistic fusion schemes that fail to reconcile both across diverse environments [3], [6]. Without systematic balancing, a binary trade-off emerges: overemphasizing coarse and uncertain semantic guidance yields circuitous routes that exhaust the battery, while favoring geometric efficiency leads to myopic exploration that overlooks critical regions. These limitations are most pronounced in expansive outdoor environments with sparse semantic cues and diverse layouts, restricting search performance.

To address these challenges, we propose `AirHunt`, which achieves smooth and efficient aerial object navigation outdoors by seamlessly bridging VLM-based semantic reasoning with continuous path planning. As shown in Fig. 1, `AirHunt` efficiently locates *open-set* objects with *zero-shot* generalization across diverse scenarios, featuring key designs in both system architecture and algorithms to fully unleash the VLM's potential, as elaborated below.

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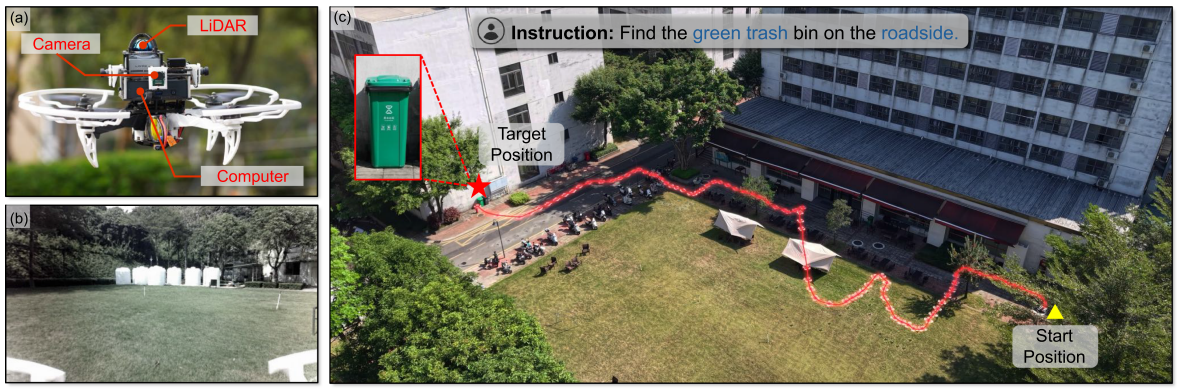


Fig. 1. A drone performs object navigation in a complex outdoor scene following a natural language instruction. (a) Custom-built aerial platform equipped with a LiDAR, an onboard computer, and three cameras. (b) Snapshot of the drone's first-person view during navigation. (c) Bird's-eye view of the experimental scenario, where the flight trajectory (red line) extends from the corner of the grassy area (yellow triangle) to the target (red star).

On the architectural front, we reposition the VLM as a high-level dense semantic generator rather than a direct action generator. Instead of emitting discrete actions, it predicts region-level target-presence probabilities from selected keyframes, yielding dense semantic priors that are fused into a 3D value map to guide motion planning. To mitigate limited 3D scene understanding, this map performs confidence-weighted, cross-view integration of these language-conditioned priors, reconciling contradictory VLM outputs over time. This 3D representation enables a *Dual-pathway Asynchronous* design in which the VLM and planner run at their native frequencies, alleviating the latency-mismatch bottleneck.

On the algorithm front, we complement the architecture with two targeted optimizations. (*i*) *Active Dual-Task Reasoning (ADTR):* while the architecture enables asynchronous inference, continuous flight produces highly redundant image streams that make per-frame VLM querying prohibitive. ADTR queries the VLM only when observations provide novel geometric coverage or instruction-relevant content, turning it into an active reasoner that maintains scene awareness while substantially reducing computation. (*ii*) *Semantic-Geometric Coherent Planning (SGCP):* SGCP reconciles the tension between semantic guidance and motion efficiency through a selective constraint injection mechanism that activates precedence constraints only between regions with significantly different semantic values, then minimizes total travel distance under these constraints. This yields a dynamic partial order that prioritizes high-value regions where semantic relevance is distinct and falls back to geometric proximity where values are comparable, adapting to semantic guidance in heterogeneous scenes and geometry-driven efficiency in homogeneous ones.

We conduct extensive experiments in the high-fidelity Unreal Engine simulator [7], covering diverse object navigation tasks across diverse outdoor scenarios. `AirHunt` achieves a 73.1% average success rate and an 11.6 m average navigation error, outperforming baselines by 49.1 percentage points and 82.1%, respectively, while reducing total flight time by 59.2%. Over ten hours of real-world experiments on a customized quadrotor further demonstrate its practical effectiveness.

In summary, this letter makes the following contributions.

- We propose `AirHunt`, an outdoor open-set aerial object navigation system that repositions VLMs as dense semantic generators rather than direct action generators and integrates semantics via a 3D value map, coupled with a dual-pathway asynchronous design that enables continuous flight with adaptive semantic guidance.
- We propose ADTR that exploits geometric and semantic redundancy for selective VLM querying; and SGCP, a unified optimization framework that dynamically reconciles semantic priorities and motion efficiency, enabling seamless adaptation to heterogeneous environments.
- We extensively evaluate our system in both simulation across diverse navigation tasks and real-world deployment on a customized quadrotor in challenging outdoor scenarios, demonstrating its zero-shot generalization.

II. Related Work

A. Object Goal Navigation

Object Goal Navigation (ObjectNav) [8] requires agents to navigate to specified object categories in unknown environments using egocentric observations. While early methods rely on task-specific training such as reinforcement learning [9] for closed-set categories, recent zero-shot approaches leverage foundation models for open-set generalization. LLM-based methods [10] exploit commonsense reasoning about object-scene relationships to evaluate exploration frontiers, whereas VLM-based methods [11], [12] directly extract semantic priors from visual observations to construct spatial value representations and guide exploration. However, existing works predominantly target small-scale, semantically structured indoor environments, where hierarchical room-object layouts provide strong localization cues. In contrast, our work extends ObjectNav to drone platforms operating in large-scale outdoor scenarios, where extended search ranges, sparse target distributions across unstructured terrain, and the absence of hierarchical semantic organization pose challenges. Though value map-based methods have proven effective indoors, transferring them to our aerial outdoor setting is nontrivial, which motivates our dual-pathway asynchronous design.

B. Aerial Object Search

Early aerial search methods rely on geometry-driven [1] or information-driven [13] exploration with closed-set detectors, lacking semantic understanding and open-set generalization. Recent VLM-based methods follow three main paradigms: (*i*) motion primitive-based selection [2], [3], where VLMs output discrete navigation commands; (*ii*) coordinate selection [6],

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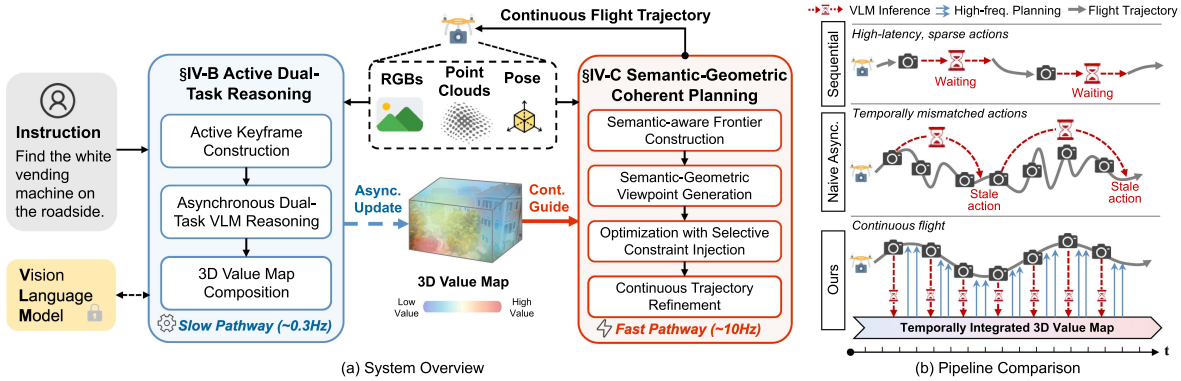


Fig. 2. **System overview and pipeline comparison.** (a) AirHunt features a dual-pathway asynchronous architecture. The VLM-driven reasoning pathway extracts language-conditioned semantic priors to asynchronously (Async.) update a 3D value map, while the high-frequency planning pathway continuously (Cont.) harvests this evolving memory to generate trajectories. Both modules operate at their native frequencies without mutual blocking. (b) Pipeline comparison. The Sequential approach forces drones to hover for VLM responses, causing fragmented flight. The Naive Asynchronous approach allows movement during inference but issues action commands based on stale observations. In contrast, AirHunt repositions the VLM as a high-level semantic generator and integrates its outputs into a persistent 3D value map, enabling continuous flight under adaptive semantic guidance.

where VLMs pick target locations from overlaid candidate coordinates; and (*iii*) coordinate prediction [14], where VLMs predict 2D pixel coordinates projected into 3D space for navigation. While effective in constrained scenarios, these paradigms suffer from stop-and-go behavior and over-reliance on VLM decisions, limiting search efficiency in expansive outdoor environments. APEX [5] adopts a hierarchical design but maps its semantic memory to actions via RL, which limits open-set generalization. In contrast, we propose a dual-pathway asynchronous architecture that decouples VLM inference from path planning. The planner prioritizes VLM-inferred high-value regions while minimizing travel distance, yielding continuous flight trajectories for efficient search.

III. Problem Formulation

This letter addresses aerial object navigation, a variant of the ObjectNav [8] task adapted for aerial agents in large-scale outdoor environments. The agent must explore an unknown environment to locate an open-set target object specified by a natural language instruction, while minimizing path length. Task success is achieved when the agent reaches within a predefined distance of the target and executes a stop action.

IV. Methodology

A. Asynchronous System Architecture

The system architecture of AirHunt is illustrated in Fig. 2(a). AirHunt adopts a *Dual-pathway Asynchronous* architecture that repurposes the VLM as a dense semantic signal generator, producing language-conditioned, region-level semantic priors for spatially dense guidance. The VLM and planner operate at their native frequencies with their synergy mediated by a shared 3D value map. The low-frequency VLM-driven reasoning pathway infers priors from selected observations and asynchronously integrates them into the map, while the high-frequency planning pathway continuously generates trajectories from the map by jointly leveraging semantic priors and geometric information without synchronizing with VLM inference. By integrating semantic evidence over time, the shared map reconciles potential contradictions in VLM outputs and mitigates VLM's limited 3D scene understanding, enabling continuous long-horizon flight with dense semantic guidance.

From the algorithm perspective, AirHunt comprises two core modules: Active Dual-Task Reasoning (ADTR, Section IV-B) updates the 3D value map and verifies targets via VLM queries on informative keyframes, while Semantic-Geometric Coherent Planning (SGCP, Section IV-C) generates spatially consistent search trajectories that balance semantic guidance with geometric efficiency. AirHunt operates in three stages: (*i*) Initialize: the drone rotates 360° in place to observe the surroundings; (*ii*) Search: ADTR continuously updates the 3D value map to guide SGCP in generating exploration trajectories; (*iii*) Navigate to target: upon target confirmation by ADTR, the system transitions to goal-directed navigation until reaching the target.

B. Active Dual-Task Reasoning

Continuous flight produces highly redundant image streams, making exhaustive per-frame VLM processing (>2000 ms per query) computationally prohibitive. We thus propose ADTR (Fig. 3), which selectively queries the VLM on observations with novel geometric coverage or instruction-relevant content.

Active Keyframe Construction: We adaptively construct two keyframe types: coverage-aware keyframes that maximize geometric exploration coverage, and task-aware keyframes that capture task-relevant observations.

1) Coverage-aware Keyframe Construction: We discretize the environment into a 3D voxel grid with resolution r. A coverage-aware keyframe is defined as $\mathcal{K}_i^{\text{cov}} = \langle I_i, \mathcal{V}(I_i) \rangle$, where i denotes the keyframe index, I_i is the observation (RGB image with camera pose) and $\mathcal{V}(I_i) \subset \mathbb{R}^3$ is the set of visible voxels. The keyframe set at time t is denoted $\mathcal{S}_t^{\text{cov}}$. To maximize coverage while minimizing redundancy, we define the overlap ratio between the candidate keyframe $\mathcal{K}_t^{\text{cov}}$ and an existing keyframe $\mathcal{K}_i^{\text{cov}}$ as $\mathcal{R}\left(\mathcal{K}_t^{\text{cov}}, \mathcal{K}_i^{\text{cov}}\right) = \left|\mathcal{V}(I_t) \cap \mathcal{V}(I_i)\right| / \left|\mathcal{V}(I_t)\right|$. The candidate is added as a new keyframe only if its overlap with all existing keyframes remains below threshold τ_{cov}:

$$\mathcal{S}_t^{\text{cov}} = \mathcal{S}_{t-1}^{\text{cov}} \cup \{\mathcal{K}_t^{\text{cov}}\}, \text{ if } \mathcal{R}\left(\mathcal{K}_t^{\text{cov}}, \mathcal{K}_i^{\text{cov}}\right) < \tau_{\text{cov}}, \quad \forall \mathcal{K}_i^{\text{cov}} \in \mathcal{S}_{t-1}^{\text{cov}}. \tag{1}$$

This retains observations with sufficient novel coverage while keeping the explored representation compact.

2) Task-aware Keyframe Construction: Task-aware keyframes capture object-level details for target verification. A task-aware keyframe $\mathcal{K}_i^{\text{task}} = \langle I_j, \mathcal{O}(I_j) \rangle$ pairs observation I_j with

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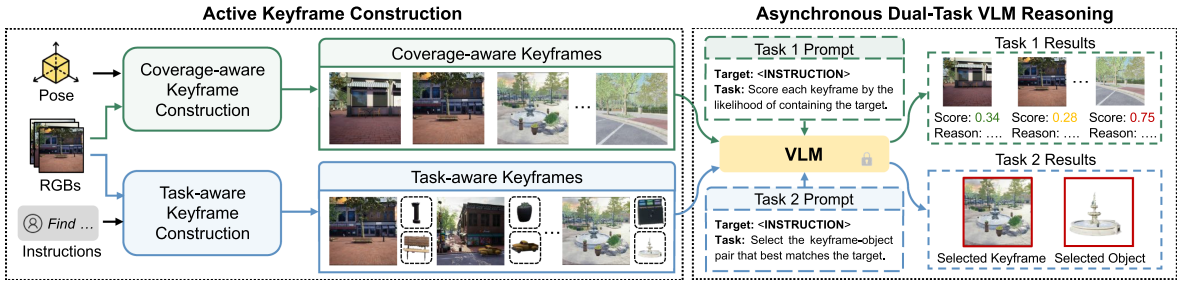


Fig. 3. Illustrations of the Active Keyframe Construction and Asynchronous Dual-Task VLM Reasoning Modules. ADTR constructs a coverage-aware keyframe set and a task-aware keyframe set, then queries the VLM for semantic value inference (Task 1) and related object verification (Task 2), respectively.

detected objects $\mathcal{O}(I_j) = \{o_1, \ldots, o_M\}$, where each $o_m = (c_m, \gamma_m, \mathbf{x}_m)$ specifies category, confidence, and 3D position. We maintain the set $\mathcal{S}_t^{\text{task}}$. Given an instruction $\mathcal{L}$, a VLM infers online the top-K target-relevant categories $\mathcal{C}_{\text{target}} = \{c_1^*, \ldots, c_K^*\}$. A candidate is added only when the current observation contains at least one target-relevant detection: $\mathcal{S}_t^{\text{task}} = \mathcal{S}_{t-1}^{\text{task}} \cup \{\mathcal{K}_t^{\text{task}}\}$, if $\mathcal{C}(I_t) \cap \mathcal{C}_{\text{target}} \neq \emptyset$. This filtering retains task-critical observations while discarding semantically irrelevant frames for VLM reasoning.

Asynchronous Dual-Task VLM Reasoning: The constructed keyframes are fed to our VLM reasoning module for semantic value inference and object verification, with both tasks executing asynchronously to hide VLM latency.

1) Task 1: Semantic Value Inference: We use the VLM to infer region-level target-presence probabilities for the scenes captured in keyframes, providing semantic priors for subsequent motion generation. Given an instruction $\mathcal{L}$ and a coverage-aware keyframe set $\mathcal{S}_t^{\text{cov}}$, we assign each keyframe $\mathcal{K}_i^{\text{cov}} \in \mathcal{S}_t^{\text{cov}}$ a semantic value $v_i \in [0,1]$ via $v_i = \text{VLM}(\mathcal{K}_i^{\text{cov}}, \mathcal{L})$, where v_i quantifies the probability that the region represented by $\mathcal{K}_i^{\text{cov}}$ contains the target. For example, given "find the football on the grass", keyframes of grassy areas receive higher values than those of basketball courts.

2) Task 2: Related Object Verification: Open-vocabulary detectors may produce erroneous labels under occlusion, distance, or visual ambiguity and fail to disambiguate semantically similar objects (e.g., a specific black trash bin among bins of various colors). We thus use the VLM to verify targets from detection candidates. Given a task-aware keyframe set $\mathcal{S}_t^{\text{task}}$ and an instruction $\mathcal{L}$, the VLM selects the target via $(\mathcal{K}_*^{\text{task}}, o^*) = \text{VLM}(\mathcal{S}_t^{\text{task}}, \mathcal{L})$, where $\mathcal{K}_*^{\text{task}}$ is the selected keyframe and $o^* \in \mathcal{O}(\mathcal{K}_*^{\text{task}})$ is the identified target among detection candidates. By jointly exploiting appearance and spatial context (e.g., relative positions and background), the VLM robustly resolves ambiguous detections.

3D Value Map Composition: To guide search, we maintain a 3D value map $\mathbf{V} \in \mathbb{R}^{W \times H \times D}$ with channels $\mathbf{V} = (\mathbf{V}^{\text{sem}}, \mathbf{V}^{\text{conf}})$, where $\mathbf{V}^{\text{sem}} \in [0,1]$ encodes target-presence potential and $\mathbf{V}^{\text{conf}} \in [0,1]$ encodes estimation confidence. The map is updated *asynchronously* whenever Task 1 returns a semantic value, enabling continuous navigation without blocking on VLM latency. Given v_i for keyframe $\mathcal{K}_i^{\text{cov}}$, we project it to voxels in $\mathcal{V}(I_i)$ excluding occluded regions. For voxel k at $\mathbf{x}_k$ observed from pose $\mathbf{p}_t$, per-observation confidence decays with distance as $\hat{c}_k^t = \max(0, 1 - \|\mathbf{x}_k - \mathbf{p}_t\|_2 / d_{\max})$, with $d_{\max}$ the maximum sensing range. Channels are fused via confidence-weighted aggregation [11] as $v_k^t = (c_k^{t-1} v_k^{t-1} + \hat{c}_k^t v_i) / (c_k^{t-1} + \hat{c}_k^t)$ and $c_k^t = \left((c_k^{t-1})^2 + (\hat{c}_k^t)^2\right) / (c_k^{t-1} + \hat{c}_k^t)$, where quadratic accumulation emphasizes higher-confidence observations. Multi-view fusion of language-conditioned evidence reconciles inconsistent VLM outputs into a coherent spatial belief and mitigates limited 3D scene understanding. In this way, each per-keyframe semantic value acts as a soft prior rather than a hard decision, so occasional noisy or miscalibrated zero-shot estimates are absorbed by the multi-view fusion instead of directly driving the search, and their residual effect is further tempered by the selective precedence constraints in planning (Section IV-C).

C. Semantic-Geometric Coherent Planning

Planning over the 3D value map is challenging because semantic priors are probabilistic: strictly prioritizing high-value regions causes oscillatory motion, while prioritizing geometric proximity delays visiting target-relevant regions. We thus propose hierarchical SGCP that fuses semantic priors with geometric costs at three levels: frontier construction, viewpoint generation, and global tour optimization.

Semantic-aware Frontier Construction: Traditional frontier-based methods rely on purely geometric metrics for coverage [15], and recent frontier-based methods [11], [12], [16] attach semantic scores to geometrically constructed 2D frontiers for selection. We instead embed semantics into the frontier construction itself, introducing a semantic frontier that integrates geometric information with semantic cues.

1) Semantic Frontier Formulation: We represent the environment with a volumetric occupancy grid [17], assigning each frontier voxel k a semantic value $v_k \in [0,1]$ from the 3D value map. A *semantic frontier cluster* is defined as $F_i = \langle \mathcal{C}_i, \mathbf{p}_i, s_i \rangle$, where $\mathcal{C}_i = \{c_1, \ldots, c_{n_i}\}$ is the frontier voxel set, $\mathbf{p}_i \in \mathbb{R}^3$ its geometric centroid, and $s_i = \frac{1}{|\mathcal{C}_i|} \sum_{c_j \in \mathcal{C}_i} v_{c_j}$ its average semantic value.

2) Incremental Frontier Update: Upon each occupancy map update, we first remove clusters affected by newly observed regions [18], then identify new frontiers via *semantic-constrained region growing*. From seed voxels c_s, we expand clusters by breadth-first expansion over 6-connected neighbors, adding $c_j \in \mathcal{N}(c_k)$ to F_i only if $|v_{c_j} - v_{c_s}| < \tau_s$ and $c_j \in \mathcal{U}$, where τ_s is a semantic similarity threshold and $\mathcal{U}$ the set of unvisited frontier voxels. Oversized clusters are recursively split following [18].

Semantic-Geometric Viewpoint Generation: For each frontier cluster F_i, we generate candidate viewpoints around its centroid $\mathbf{p}_i$ to observe unexplored regions [19], yielding $\mathcal{U}_i = \left\{u_i^1, \ldots, u_i^{M_i}\right\}$, where each $u_i^j = \left(\mathbf{p}_i^j, \theta_i^j\right)$ specifies position and yaw. For each candidate, we define $\mathcal{N}_{obs}(u_i^j)$ as the unoccluded frontier voxels in its field of view (FoV)

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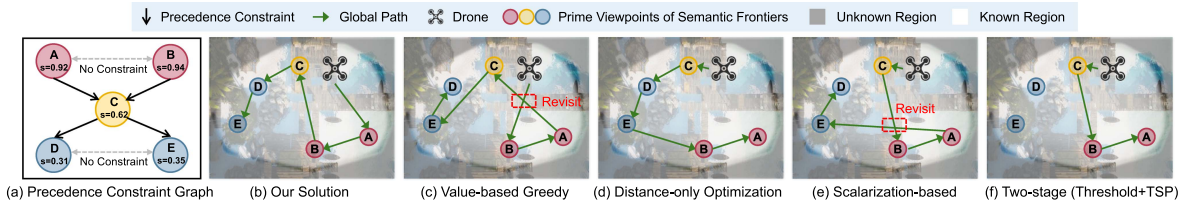


Fig. 4. Comparison of global planning strategies. (a) The constraint graph selectively enforces precedence only when semantic value differences are significant (e.g., A(0.92)→C(0.62)), allowing geometric optimization for similar-value frontiers (e.g., A and B). (b) Our solution generates a globally coherent path that prioritizes high-value frontiers without revisits. (c)-(f) Baseline approaches tend to yield revisits of previously searched regions (red dashed boxes) or delay visits to semantically critical regions (A and B).

and quantify semantic-weighted information gain as $S(u_i^j) = \sum_{c_k \in \mathcal{N}_{obs}(u_i^j)} v(c_k)$. The candidate with the highest information gain is selected as the *prime viewpoint* u_i^*, jointly promoting broad frontier coverage and exploration of semantically informative regions.

Optimization with Selective Constraint Injection: Given a set of semantic frontier clusters, our objective is to compute a global exploration tour that prioritizes target-relevant regions while minimizing total traversal time. Value-based greedy selection (Fig. 4(c)) incurs large detours when high-value frontiers are dispersed, whereas distance-only optimization (Fig. 4(d)) favors short moves and may defer semantically critical regions. Hybrid heuristics either (*i*) scalarize value and distance into a single utility, biasing toward nearby moderate-value frontiers (Fig. 4(e)), or (*ii*) threshold semantic values to restrict the frontier set and optimize distance within it, discarding useful regions at the hard cutoff (Fig. 4(f)).

Our key idea is to *selectively* inject ordering constraints only between clusters whose semantic values differ significantly (Fig. 4(a)), and optimize the visitation order by minimizing geometric traversal cost subject to these constraints. This design adapts to the semantic distribution: dense precedence relations enforce early visitation of distinctly target-relevant regions, while sparse constraints allow geometric proximity to dominate when scores are comparable. Through value-gap-derived precedence constraints, SGCP adaptively balances semantic-driven exploration and geometry-efficient routing without environment-specific parameter tuning. The formulation can be transformed into a Sequential Ordering Problem (SOP) and effectively solved by existing solvers.

Definition 1: (Sequential Ordering Problem) [20]: Given a directed graph $G = (V, E)$, edge weight $w : E \to \mathbb{R}$ and a set of precedence constraints $\mathcal{P} \subseteq V \times V$, the SOP searches for a minimum cost permutation of vertices from $s \in V$ to $t \in V$ that satisfies the precedence constraints.

1) Selective Precedence Constraint: Each cluster F_i is mapped to a vertex $i \in V$ at its prime viewpoint u_i^*, and precedence constraints are induced by semantic value gaps as $\mathcal{P} = \{(i, j) \mid s_i - s_j > \rho\}$, where s_i is the semantic value of F_i and ρ is a sensitivity threshold, requiring u_i^* to be visited before u_j^*. Only semantically significant gaps thus trigger ordering constraints (Fig. 4(a)), leaving geometric optimization to dominate when semantic cues are weak.

2) Geometric Cost Calculation: For viewpoints $u_i = (\mathbf{p}_i, \theta_i)$ and $u_j = (\mathbf{p}_j, \theta_j)$, the geometric cost (edge weight w) is

$$C(u_i, u_j) = \max \left\{ \frac{d(\mathbf{p}_i, \mathbf{p}_j)}{v_{\max}}, \frac{|\theta_i - \theta_j|}{\omega_{\max}} \right\}, \tag{2}$$

where $d(\mathbf{p}_i, \mathbf{p}_j)$ is the path length computed via A* algorithm, and $v_{\max}$, $\omega_{\max}$ are the maximum translational and angular velocities. Inter-cluster cost is defined between prime viewpoints as $C_{\text{geo}}(F_i, F_j) = C(u_i^*, u_j^*)$.

3) Cost Matrix Integration: We construct a unified cost matrix $\mathbf{C}_G \in \mathbb{R}^{(N_c+1)\times(N_c+1)}$ integrating geometric costs and semantic constraints:

$$\mathbf{C}_G = \begin{bmatrix} c_{c,c} & \mathbf{c}_{c,k}^T \\ \mathbf{c}_{k,c} & \mathbf{C}_{k,k} \end{bmatrix}, \tag{3}$$

where subscript c and k denote the current position and frontier clusters, respectively. The inter-cluster block $\mathbf{C}_{k,k} \in \mathbb{R}^{N_c \times N_c}$ is

$$\mathbf{C}_{k,k}(i, j) = \begin{cases} -1 & \text{if } (i, j) \in \mathcal{P} \text{ and } i \neq j, \\ 0 & \text{if } i = j, \\ C_{\text{geo}}(F_i, F_j) & \text{otherwise.} \end{cases} \tag{4}$$

This unified cost matrix enables the solver to enforce semantic priorities through $\mathcal{P}$ while minimizing geometric travel cost where no constraints apply. The vector $\mathbf{c}_{c,k} \in \mathbb{R}^{N_c}$ stores costs from the current position to each cluster via (2), and setting $\mathbf{c}_{k,c} = -\mathbf{1}_{N_c}$ with $c_{c,c} = 0$ anchors the tour at the current position.

Continuous Trajectory Refinement: Once the global cluster visitation order is fixed, we refine local viewpoint selection between consecutive frontiers to minimize execution time [18]. After ADTR confirms the target, we instead select a line-of-sight viewpoint near the verified target. A B-spline trajectory is then generated to jointly optimize smoothness and execution time under safety and dynamic feasibility constraints [21], with real-time replanning when new frontiers appear or the 3D value map updates.

V. Experiments

A. Experimental Setup

Simulation Environments: We evaluate in Unreal Engine 4.27 [7] on four large-scale outdoor environments (e.g., Downtown, Cabinlake) spanning 160,000–562,500 m^2 across villages, cities, forests, roads, seas, and mountains. The benchmark comprises 85 episodes over 25 object categories with open-ended, environment-aware instructions (e.g., Find the boat on water). Environments, the instruction word cloud, and representative objects are shown in Fig. 5.

Metrics: Following prior works [2], [3], we adopt five metrics: Success Rate (SR), Oracle Success Rate (OSR), Success weighted by inverse Path Length (SPL), Distance to Goal (DTG), and Flight Time (FT). SR and OSR measure whether the drone stops or ever comes within a threshold distance of the goal, respectively. SPL weights success by the ratio of optimal to actual path length. DTG records the final Euclidean distance to

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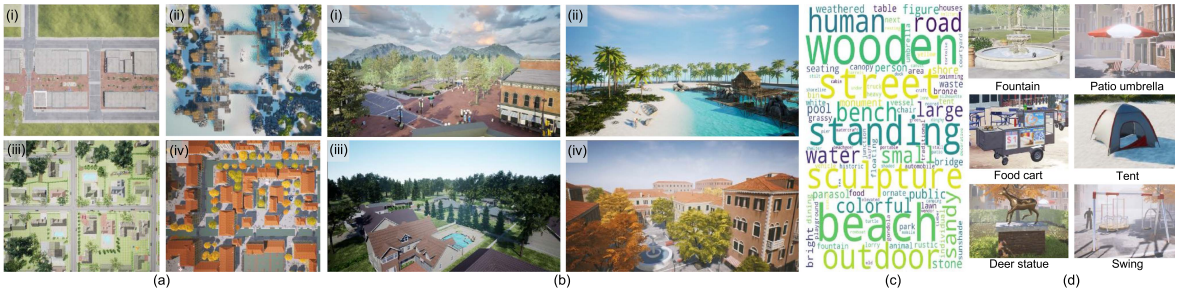


Fig. 5. (a) Top-down views and (b) snapshots of four simulation environments: (i) Downtown, (ii) Cabinlake, (iii) Neighborhood, and (iv) Venice. (c) Word cloud of task instructions. (d) Images of example target objects.

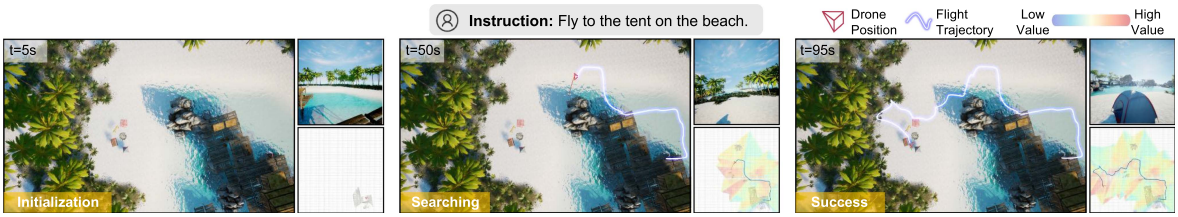


Fig. 6. Example qualitative results of AirHunt in the Cabinlake environment, showing the top-down view (left), drone's first-person view (top right), and 3D value map (bottom right). AirHunt prioritizes regions likely containing the target and avoids revisits, demonstrating spatial consistency.

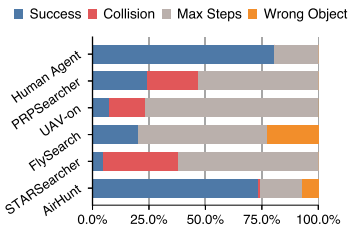


Fig. 7. Failure breakdown of AirHunt and baseline methods across four categories in simulation experiments.

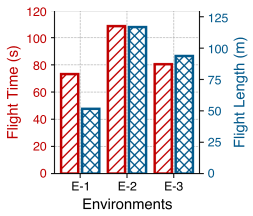


Fig. 8. Total flight time and flight length in three real-world experimental tasks.

the target, providing a continuous measure of residual localization error beyond the binary thresholded SR/OSR outcomes. FT measures total flight time per episode including VLM inference.

Baselines: PRPSearcher [2]: a motion primitive-based pipeline with a semantic attraction map and VLM-driven discrete action selection. *UAV-on* [3]: converts RGB, depth, and pose into structured text, then queries an LLM for discrete navigation commands. *FlySearch* [6]: overlays RGB with a coordinate grid and queries a VLM to pick the next target location. *STARSearcher* [1]: a geometric method alternating between unknown-area exploration and surface inspection. *APEX* [5]: builds a VLM-based spatio-semantic memory and maps it to actions via RL. *Human Agent:* mean performance of three expert operators with 20+ hours of simulator training.

TABLE I
OVERALL PERFORMANCE COMPARISON. BEST RESULTS IN CYAN, SECOND-BEST IN VIOLET, AND HUMAN AGENT IN GRAY.

Method	SR ↑	OSR ↑	SPL ↑	DTG ↓	FT ↓
Human Agent	80.5	81.7	61.7	18.8	24.6
PRPSearcher	24.0	26.6	19.3	90.2	324.6
UAV-on	7.4	33.3	3.8	64.9	364.1
FlySearch	20.3	34.2	8.6	137.5	767.7
STARSearcher	2.4	4.9	1.1	103.5	296.7
APEX	15.2	17.6	8.3	51.4	193.1
Sequential	48.2	52.9	24.8	25.7	211.5
Naive-Async	68.4	73.7	37.0	13.0	187.6
AirHunt (Ours)	**73.1**	**80.5**	**42.0**	**11.6**	**120.8**

Implementation Details: We simulate a drone in AirSim [22] with a 10 m LiDAR and three RGB cameras (front/left/right, each $90^\circ \times 60^\circ$ FoV), subject to $v_{\max} = 2.0$ m/s, $\mathbf{a}_{\max} = 1.5$ m/s^2, and $\omega_{\max} = 0.9$ rad/s. We adopt Grounding-DINO [23] for detection, SAM2 [24] for segmentation, and Qwen-VL-Max [25] as the VLM for all methods. Each detected object's 3D position is obtained by back-projecting SAM2 mask pixels with depth and camera pose and taking their centroid. We set $\rho = 0.15$, use a 5 m threshold for SR and OSR, and cap each task at 50 VLM queries. We set the map resolution as $r = 0.2$ m. Each method is evaluated over three runs on an Intel Core i7-14700KF with an RTX 4070 (12 GB) and 32 GB RAM. The optimization in Section IV-C is solved via [26].

B. Benchmark Results

As shown in Table I, AirHunt outperforms all baselines except the Human Agent, raising SR by 49.1 percentage points over PRPSearcher, the strongest baseline. Among baselines, STARSearcher gets trapped inspecting voxel surfaces in structurally complex regions, while UAV-on lacks active exploration, leading to poor SR. AirHunt also achieves the lowest DTG across all methods, even surpassing the Human Agent,

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Fig. 9. Real-world experiments are conducted in three challenging environments, including a Park Area (E-1), a Landscaped Area (E-2), and a Residential Area (E-3). For each task, the first image illustrates a top-down view of the experimental scenario, and the others are snapshots of different drone states during task execution. The yellow circle marks the drone, and the red rectangles mark the target objects.

TABLE II
ABLATION STUDY ON KEY MODULES

Method	SR ↑	OSR ↑	SPL ↑	DTG ↓	FT ↓
RecTask1	59.3	68.8	40.7	25.6	128.7
RecTask2	31.2	38.4	12.5	65.1	135.6
Value-greedy	16.2	18.3	5.5	105.6	312.7
Distance-only	42.4	45.9	27.4	25.7	104.2
Scalarization	33.2	38.9	17.5	54.2	239.4
Two-stage	58.9	63.4	32.5	21.4	198.2
AirHunt	**73.1**	**80.5**	**42.0**	**11.6**	**120.8**

TABLE III
AVERAGE COMPUTATION TIME (MS) OF KEY COMPONENTS

ADTR (§IV-B)			SGCP (§IV-C)		
Task 1	Task 2	Value Map	Front. +VP.	Global Path	Traj. Refine
3616	4555	8	2.3	20.7	0.4

by leveraging the target's 3D position from ADTR's task-aware keyframes for precise stopping. For flight efficiency, PRPSearcher requires an average FT that is 2.5× longer than AirHunt, as its stop-and-go behavior disrupts flight continuity, whereas our asynchronous architecture decouples VLM reasoning from planning to enable smoother flight. APEX [5] attains the best DTG and FT among external baselines but only 15.2% SR, as its RL action policy requires task-specific training and generalizes poorly in our open-set setting. A representative episode is illustrated in Fig. 6. Table I further ablates the three architectures of Fig. 2, confirming that resolving the frequency mismatch requires our dual-pathway design rather than asynchronous execution alone.

C. Ablation Study

Ablation Study on ADTR:. We evaluate ADTR with two variants in Table II: (*i*) *RecTask1* replaces coverage-aware keyframes with the most recent egocentric observations for Task 1 VLM queries; (*ii*) *RecTask2* directly feeds raw observations to the VLM for target detection and distance estimation. Other components follow AirHunt. RecTask1 drops SR from 73.1% to 59.3%, as recent observations lack spatial coverage for VLM reasoning. RecTask2 degrades further since VLMs cannot reliably estimate target distance from raw images, causing premature termination under occlusion. These results confirm that ADTR provides compact and informative content for effective VLM reasoning.

Ablation Study on SGCP: We compare SGCP against four planning variants: (*i*) *Value-greedy* always picks the highest-value frontier; (*ii*) *Distance-only* minimizes geometric travel cost; (*iii*) *Scalarization* maximizes a value/distance utility; and (*iv*) *Two-stage* [12] filters frontiers by a value threshold, then solves a TSP tour over the subset. As shown in Table II, all variants degrade on SR. Value-greedy suffers the most (SR drops by 56.9 percentage points) as its myopic decisions trigger revisits, while Scalarization underperforms by deferring distant but higher-value regions. Distance-only achieves the shortest flight time, even below AirHunt, but its SR remains far lower as it defers semantically critical regions (Fig. 4(d)). Two-stage is the strongest variant, yet its hard threshold may discard target-containing regions. In contrast, AirHunt prioritizes higher-value frontiers via precedence constraints while ordering similarly-valued ones by proximity, balancing semantic priority and travel efficiency.

D. Computation Time

We evaluate the computation time of key components, measured per update for ADTR and per planning iteration for SGCP. As shown in Table III, the planning pipeline is lightweight, with even the most expensive step (global tour optimization, Section IV-C) averaging about 21 ms per iteration. Consequently, the onboard planner sustains ∼10 Hz, whereas VLM queries remain sparse at only ∼0.3–0.4 Hz. The two pathways thus run concurrently without mutual blocking, confirming the

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necessity of the asynchronous decoupling of planning from VLM reasoning.

E. Failure Breakdown

We analyze failure reasons across all methods in Fig. 7, categorizing outcomes into *Success*, *Collision*, *Max Steps* (exceeding the VLM query limit), and *Wrong Object*. Three observations emerge: (i) Baselines mainly fail by exceeding max steps due to inefficient VLM-reliant exploration; (ii) Most baselines show high collision rates: STARSearcher collides while inspecting surfaces at close range, while others suffer from discrete motion primitives and lack of replanning. `AirHunt` substantially reduces collisions via continuous replanning, with occasional failures only in extremely narrow regions like dense forests; (iii) Our system inherits limitations from its detector, Grounding-DINO [23], which may hallucinate or misclassify background elements (e.g., shadows) under texture ambiguity or lighting variation. The resulting errors surface as Wrong Object, where the drone stops at an incorrect candidate during final verification. Since baselines mostly fail earlier at Max Steps or Collision and rarely reach the target, this mode appears mainly for `AirHunt`. It is therefore a byproduct of effective exploration rather than weaker perception.

F. Real-World Experiments

We conduct field experiments on a customized quadrotor equipped with an Intel NUC 11 Pro (8 GB), a Mid360 LiDAR, and three OAK-4P-New cameras (facing forward, left, right). FAST-LIO [27] handles onboard localization and mapping, while a ground station laptop queries the online VLM via 4 G cellular network. As shown in Fig. 9, we evaluate `AirHunt` on three open-set tasks in challenging outdoor environments: **E-1** locating a black sedan from a complex park intersection; **E-2** locating a white chair from beside a barrier in a landscaped area; and **E-3** locating a black trash bin from a grassland corner in a residential zone. As shown in Fig. 8, the system completes all tasks in 73.2 s, 108.5 s, and 80.5 s without collision. Across these flights, `AirHunt` issued on average 33 VLM queries per task, with an average VLM latency of 2.23 s and a replanning frequency of 12.3 Hz. Detailed results are provided in the Supplementary Material.

VI. Conclusion

This letter introduces `AirHunt`, an aerial object navigation system that efficiently locates open-set objects with zero-shot generalization to diverse outdoor scenarios. At its core, an asynchronous architecture enables continuous flight while incorporating intermittent VLM guidance. Two synergistic modules, *Active Dual-Task Reasoning* and *Semantic-Geometric Coherent Planning*, further enhance search efficiency and consistency. Extensive evaluations in simulation and real-world scenarios demonstrate its superior performance. While `AirHunt` performs well, it has several limitations worth exploring. First, scaling to city-scale tasks introduces further challenges such as memory usage, which we plan to address through hierarchical map representations. Second, a flight-time gap to expert operators remains, as episode time includes repeated ADTR queries and zero-shot inference is less decisive than expert search. We plan to narrow it by consolidating Task 1 and Task 2 into a unified semantic module, fine-tuning on expert trajectories, and improving VLM inference efficiency.

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